

WEB PROGRAMMING LAB-9 HOME EXERCISES

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Title: Form Processing using Django – Part II

Question 1

Create a website with two pages.

First page contains:

RadioButton with HP, Nokia, Samsung, Motorola, Apple as options.

CheckBox with Mobile and Laptop as items.

TextBox to enter quantity.

There is a button with text as "Produce Bill".

On Clicking Produce Bill button, item should be displayed with total amount on another page.

Create project & App

```
# Create Django project
django-admin startproject bill_project
cd bill_project

# Create app
python manage.py startapp billapp
```

bill_project/settings.py

```
import os # added at the top

INSTALLED_APPS = [
    'billapp',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]

STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static']
```

Database Initialization

To initialize the database and create required tables, executed:

```
python manage.py migrate
```

Project URLs: `bill_project/urls.py`

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('billapp.urls')),
]
```

App URLs: `billapp/urls.py`

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.index, name='index'),
    path('generate_bill/', views.generate_bill, name='generate_bill'),
]
```

View: `billapp/views.py`

```
from django.shortcuts import render

def index(request):
    return render(request, 'index.html')

def generate_bill(request):
    if request.method == 'POST':
        brand = request.POST.get('brand')
        # getlist is used for multiple checkboxes
        items = request.POST.getlist('items')
        quantity = int(request.POST.get('quantity', 1))

        # Simple Price Logic (Base prices)
        brand_prices = {
            'HP': 500, 'Nokia': 200, 'Samsung': 400,
            'Motorola': 300, 'Apple': 800
        }
        item_multipliers = {'Mobile': 1, 'Laptop': 2.5}

        base_price = brand_prices.get(brand, 0)
        total_amount = 0
        selected_items_summary = []

        for item in items:
            item_price = base_price * item_multipliers.get(item, 1)
```

```

        total_amount += item_price * quantity
        selected_items_summary.append(f"{brand} {item}")

    context = {
        'brand': brand,
        'items': ", ".join(selected_items_summary),
        'quantity': quantity,
        'total': total_amount
    }
    return render(request, 'bill.html', context)

return render(request, 'index.html')

```

Create template/ in billapp/. Inside template/

bill.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Final Bill</title>
</head>
<body style="background-color: #f8f9fa; font-family: sans-serif;">
    <center>
        <div style="background: white; width: 450px; padding: 40px; margin-top: 60px; border: 2px dashed #333; text-align: center;">
            <h1 style="margin-top: 0;">INVOICE</h1>
            <hr>
            <table width="100%" cellpadding="10" style="text-align: left;">
                <tr>
                    <td><strong>Items Purchased:</strong></td>
                    <td>{{ items }}</td>
                </tr>
                <tr>
                    <td><strong>Quantity:</strong></td>
                    <td>{{ quantity }}</td>
                </tr>
                <tr>
                    <td><h2 style="color: #d9534f;">Total Amount:</h2></td>
                    <td><h2 style="color: #d9534f;">${{ total }}</h2></td>
                </tr>
            </table>
            <br>
            <a href="/" style="text-decoration: none; color: #555;"><- Create New Bill</a>
        </div>
    </center>
</body>
</html>

```

index.html

```

<!DOCTYPE html>
<html>

```

```

<head>
  <title>Produce Bill</title>
</head>
<body style="background-color: #f8f9fa; font-family: sans-serif;">
  <center>
    <div style="background: white; width: 500px; padding: 30px; margin-top: 40px; border: 1px solid #ddd; text-align: left; border-radius: 8px;">
      <h2 style="border-bottom: 2px solid #333; padding-bottom: 10px;">Store Billing</h2>

      <form action="/generate_bill/" method="POST">
        {% csrf_token %}

        <table width="100%" cellpadding="10">
          <tr>
            <td valign="top"><b>Select Brand:</b></td>
            <td>
              <input type="radio" name="brand" value="HP" checked> HP <br>
              <input type="radio" name="brand" value="Nokia"> Nokia <br>
              <input type="radio" name="brand" value="Samsung"> Samsung <br>
              <input type="radio" name="brand" value="Motorola"> Motorola <br>
              <input type="radio" name="brand" value="Apple"> Apple
            </td>
          </tr>

          <tr>
            <td valign="top"><b>Select Items:</b></td>
            <td>
              <input type="checkbox" name="items" value="Mobile"> Mobile <br>
              <input type="checkbox" name="items" value="Laptop"> Laptop
            </td>
          </tr>

          <tr>
            <td><b>Quantity:</b></td>
            <td><input type="number" name="quantity" min="1" required style="width: 80px;"></td>
          </tr>
        </table>

        <br>
        <center>
          <input type="submit" value="Produce Bill" style="padding: 8px 30px; cursor: pointer; background: #333; color: white; border: none; border-radius: 4px;">
        </center>
      </form>
    </div>
  </center>
</body>
</html>

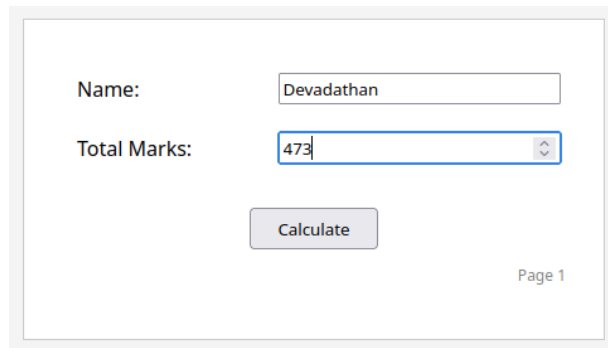
```

Run

```
python manage.py runserver 8080
```

Output

bill.html

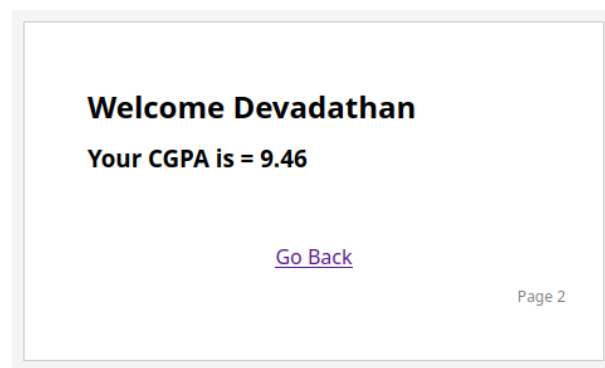


Name:

Total Marks:

Page 1

index.html



Welcome Devadathan

Your CGPA is = 9.46

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Page 2

Question 2

Write a program to display the following feedback form. The different options for the list box must be ASP-XML, DotNET, JavaPro and Unix,C,C++. When the Submit Form button is clicked after entering the data, a message as seen in the last line of the above figure must be displayed.

Create project & App

```
# Create Django project
django-admin startproject feedback-project
cd feedback-project
```

```
# Create app
python manage.py startapp feedbackapp
```

feedback_project/settings.py

```
import os

INSTALLED_APPS = [
```

```

    'feedbackapp',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]

STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static']

```

Database Initialization

To initialize the database and create required tables, executed:

```
python manage.py migrate
```

Project URLs: `feedback_project/urls.py`

```

from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('feedbackapp.urls')),
]

```

App URLs: `feedbackapp/urls.py`

```

from django.urls import path
from . import views

urlpatterns = [
    path('', views.feedback_view, name='feedback'),
]

```

View: `feedbackapp/views.py`

```

from django.shortcuts import render

def feedback_view(request):
    context = {}
    if request.method == 'POST':
        name = request.POST.get('name')
        sex = request.POST.get('sex')

        # Logic to match the "Thanks Miss. Niveditha" format
        salutation = "Mr." if sex == "Male" else "Miss."

    context = {

```

```

        'submitted': True,
        'message': f"Thanks {salutation} {name} for your feedback."
    }

    return render(request, 'feedback.html', context)

```

Create template/ in feedbackapp/. Inside template/

feedback.html

```

<!DOCTYPE html>
<html>
<head>
    <title>Courseware Feedback Form</title>
</head>
<body style="background-color: #fcfcfc; font-family: 'Times New Roman', Times, serif;">

    <div style="width: 700px; margin: 20px; border: 1px solid #ddd; padding: 20px; background-color: white;">

        <h3 style="color: brown; border-bottom: 1px solid brown; padding-bottom: 5px;">Courseware Feedback Form</h3>

        <form method="POST">
            {% csrf_token %}
            <table width="100%" cellpadding="5" border="0">
                <tr>
                    <td width="150" style="color: brown;">Student name:</td>
                    <td><input type="text" name="name" required style="width: 250px;"></td>
                </tr>

                <tr>
                    <td style="color: brown;" valign="top">Sex:</td>
                    <td>
                        <input type="radio" name="sex" value="Male" checked> Male <br>
                        <input type="radio" name="sex" value="Female"> Female
                    </td>
                </tr>

                <tr>
                    <td style="color: brown;">Select course:</td>
                    <td>
                        <select name="course" style="width: 100px;">
                            <option value="ASP-XML">ASP-XML</option>
                            <option value="DotNET">DotNET</option>
                            <option value="JavaPro">JavaPro</option>
                            <option value="Unix">Unix</option>
                            <option value="C">C</option>
                            <option value="C++">C++</option>
                        </select>
                    </td>
                </tr>

                <tr>

```

```

        <td style="color: brown;" valign="top">Technical Coverage:</td>
        <td>
            <input type="radio" name="coverage" value="Excellent" checked> Excellent
        <br>
            <input type="radio" name="coverage" value="Good"> Good <br>
            <input type="radio" name="coverage" value="Average"> Average <br>
            <input type="radio" name="coverage" value="Poor"> Poor
        </td>
    </tr>

    <tr>
        <td style="color: brown;" valign="top">Suggestions:</td>
        <td>
            <textarea name="suggestions" rows="4" style="width: 90%;"></textarea>
        </td>
    </tr>

    <tr>
        <td></td>
        <td>
            <input type="submit" value="Submit Form">
        </td>
    </tr>
</table>
</form>

{% if submitted %}
<div style="margin-top: 20px; color: brown; font-style: italic;">
    {{ message }}
</div>
{% endif %}
</div>

</body>
</html>

```

Run

```
python manage.py runserver 8080
```

Output

feedback.html

Courseware Feedback Form

Student name:

Sex: ☒ Male
☐ Female

Select course:

Technical Coverage: ☒ Excellent
☐ Good
☐ Average
☐ Poor

Suggestions:

Thanks Mr. Devadathan N R for your feedback.